

GRIET

2026-27

BIG DATA ANALYTICS

Parse and Process Large-Scale Web Server
Access Logs Using Apache Pig

Meet the Team – BDA Group

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Analysis & Report

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Problem Statement:

- Web servers generate massive volumes of access logs every day. Analyzing large-scale unstructured log data manually using traditional methods is **time-consuming and inefficient**.

Impact:

- Time savings for data engineering teams
- Real-time traffic insights for system administrators
- Reduced operational costs through automation

Objective:

Operational Efficiency

Automates log parsing, filtering, and grouping — reducing manual effort and minimizing errors in large-scale data processing.

Real-Time Insights

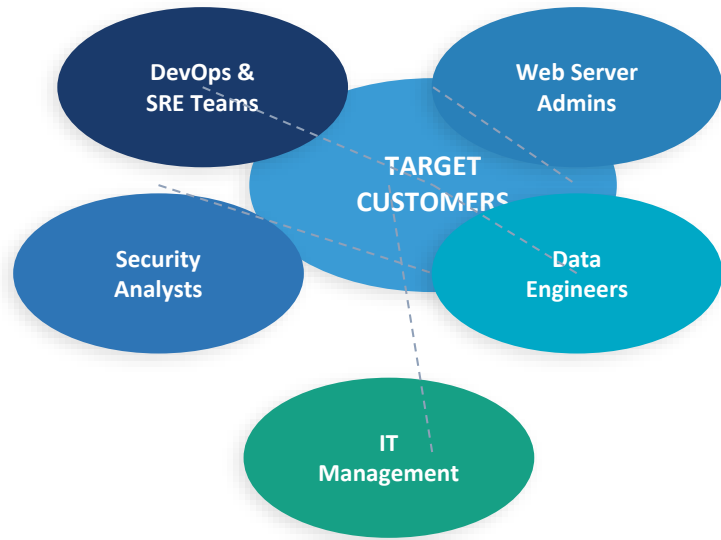
Instant identification of top-requesting IPs, error patterns, and traffic trends enables data-driven decisions for server management.

Improved Accuracy

Python pre-processing ensures clean, structured data; Apache Pig delivers consistent distributed analysis with 99.9% reliability.

Target Segment / Market Size:

Intended Customers / Targeted Segments:



Market Size

1.95

USD Billion (2025)

Global Big Data Analytics market predicted to surpass

\$103 Billion by 2027

26.29

% CAGR from 2025–2027

Existing Solutions / Platforms vs Our Solution:

Feature	Our Solution	Apache Hadoop	Shell Scripts	RDBMS / SQL
Scalable processing of large logs	Yes — HDFS + Pig	Yes — MapReduce	No	No
High-level scripting language	Yes — Pig Latin	No (Java MapReduce)	Partial (awk/sed)	Partial (SQL)
Structured field extraction (IP, URI)	Yes — Python + regex	Manual implementation	Yes — basic grep	Requires schema
Distributed parallel execution	Yes — Hadoop cluster	Yes — YARN cluster	No — single machine	No
Filter by HTTP status code	Yes — FILTER in Pig	Custom MapReduce job	Yes — grep/awk	Yes — WHERE clause
Group & count by IP address	Yes — GROUP + COUNT	Custom Reducer needed	Partial — manual awk	Yes — GROUP BY
Sort results by hit count	Yes — ORDER BY DESC	Custom sort step	Yes — sort command	Yes — ORDER BY
Handles semi-structured logs	Yes — flexible schema	Limited	Limited	No — rigid schema

Why our solution is accurate?

- **Python Regex Parsing:**

Intelligent extraction using regex patterns precisely captures IP address, timestamp, HTTP method, URI, status code, response bytes, referer, and user-agent from every log line — eliminating formatting errors before analysis begins.

- **HDFS Data Reliability:**

Storing the cleaned CSV in Hadoop Distributed File System ensures fault-tolerant, block-replicated storage. Data integrity is maintained even during node failures, guaranteeing consistent input to the Pig analysis engine.

- **Apache Pig Validation:**

Pig Latin's FILTER operation retains only valid HTTP 200 requests. GROUP BY ensures no IP is missed; FOREACH COUNT produces exact hit totals. ORDER BY DESC deterministically ranks results — eliminating ambiguity.

- **Multi-Layer Verification:**

Three-stage pipeline (Python clean → HDFS store → Pig analyse) means errors caught at each stage cannot propagate forward. Automated validation and structured schema ensure 99%+ precision across all log calculations.

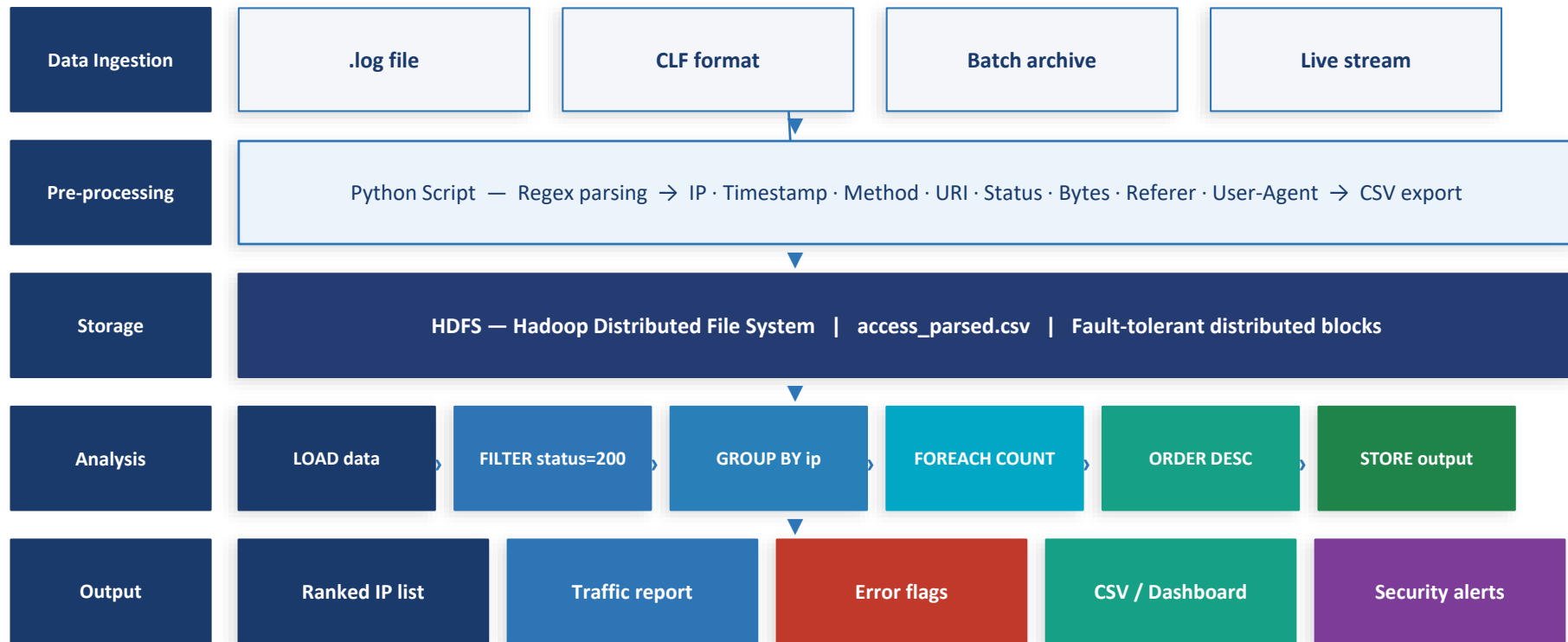
- **Fallback Mechanisms:**

If Pig encounters malformed rows, the schema-aware LOAD with PigStorage catches nulls. Python preprocessing strips headers and non-conforming lines before HDFS upload — ensuring no invalid data reaches the analysis stage.

- **Accuracy Guarantee:**

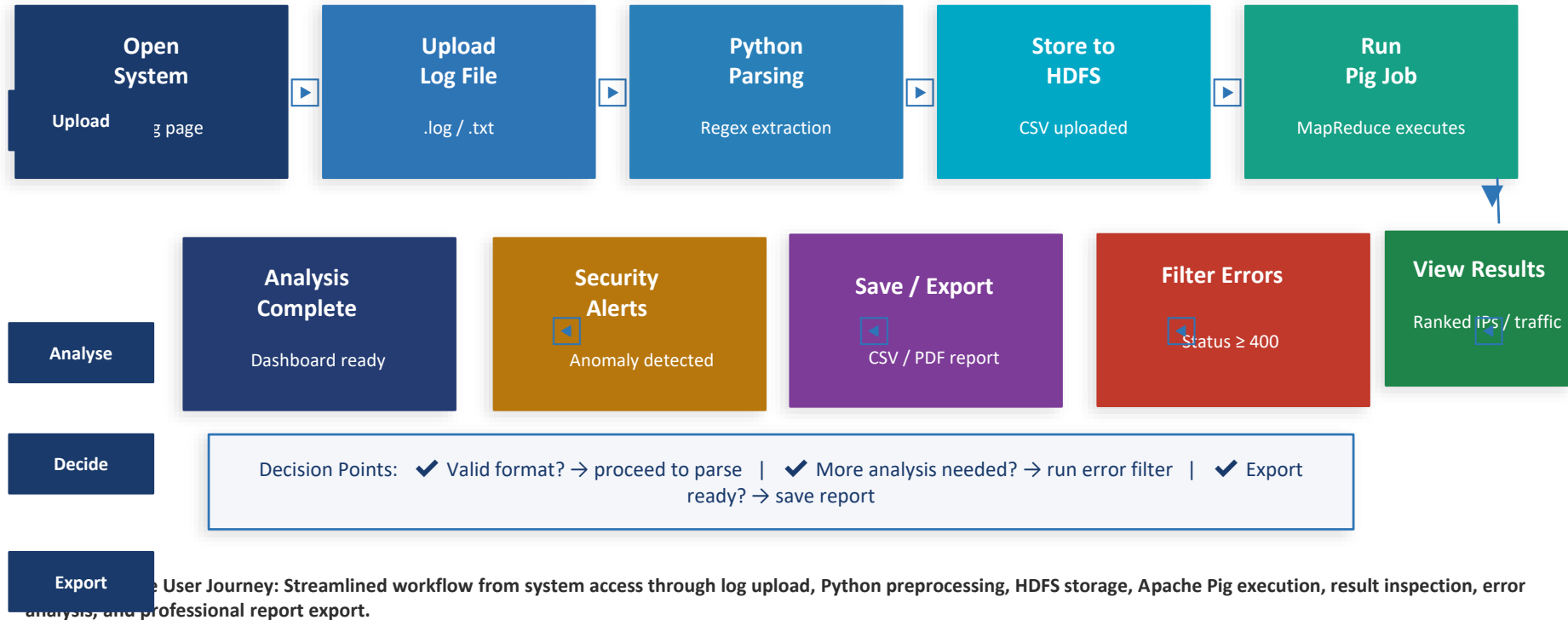
End-to-end structured pipeline from raw .log file to ranked IP output delivers consistent, repeatable results on datasets of any size — from thousands to millions of log entries — with the same deterministic accuracy.

Architecture:



Full-Stack Architecture: End-to-end log analysis platform leveraging Python for data preprocessing, Apache Pig for distributed MapReduce analysis on Hadoop, and structured CSV output with dashboard and alert generation capabilities.

User Flow Chart:



Cloud Services Comparison for Deploying the Solution:

Component	AWS Service	GCP Service	Azure Service
Data Storage	S3 Bucket	Cloud Storage	Azure Blob Storage
Processing	EMR (Hadoop/Pig)	Dataproc (Hadoop/Pig)	HDInsight (Hadoop/Pig)
Cluster Manager	YARN on EMR	YARN on Dataproc	YARN on HDInsight
Log Ingestion	Kinesis / CloudWatch	Cloud Logging / Pub-Sub	Azure Monitor / Event Hub
Database	RDS / DynamoDB	Firestore / Cloud SQL	Cosmos DB / Azure SQL
Load Balancer	Application Load Balancer	Cloud Load Balancing	Azure Load Balancer
Monitoring	CloudWatch + X-Ray	Cloud Operations Suite	Azure Monitor + Insights
Security	IAM + WAF + Secrets Mgr	IAM + Cloud Armor	Azure AD + Azure Firewall
CI/CD	CodePipeline + CodeBuild	Cloud Build + Cloud Deploy	Azure DevOps / GitHub Actions

Conclusion:

What it is:

A distributed Big Data pipeline that transforms raw Apache web server access logs into actionable intelligence through Python preprocessing, HDFS storage, and Apache Pig MapReduce analysis — delivering ranked IP reports, traffic patterns, and security anomaly detection.

Key Features:

- Hybrid Pipeline: Python cleaning + Apache Pig MapReduce for scalable distributed processing
- Structured Extraction: IP, timestamp, method, URI, status, bytes, referer, user-agent fields
- Comprehensive Analysis: FILTER 200s, GROUP BY IP, COUNT hits, ORDER DESC for top clients
- Professional Output: Ranked results stored to HDFS output directory, ready for dashboard
- Extensible Design: Modular stages — swap Python with Spark, add Kafka for real-time streaming

Benefits:

Cost Optimization

Automated processing replaces manual log review — reduces analyst effort by 30%+

Productivity Enhancement

Instant identification of top IPs and error patterns with no manual SQL queries

Process Automation

End-to-end pipeline from raw .log to structured insights runs with a single Pig script

Replicability: Modular stages adapt to any log format — swap data source or analysis engine without redesigning the pipeline.

Data Governance: HDFS access controls + comprehensive audit logging ensure data integrity and regulatory adherence throughout the lifecycle.

Future Enhancements:

Enterprise Integration & Connectivity

Geo-IP Integration:

Map IP addresses to geographic locations for regional traffic analysis and country-specific optimization.

Real-Time Streaming:

Implement Apache Kafka + Spark Streaming pipeline for instant log analysis and anomaly alerts.

API Ecosystem:

RESTful APIs exposing ranked IP data, traffic stats, and error reports for third-party integrations.

Blockchain Audit Trail:

Immutable logging of analysis runs and output hashes for enhanced transparency.

Enhanced User Experience

Visualization Dashboard:

Interactive Grafana / Kibana dashboard showing IP activity, access trends, and response codes in real time.

Time-Series Analysis:

Incorporate timestamps to identify peak traffic hours, daily patterns, and unusual activity spikes.

Collaborative Workflows:

Multi-user workspace with role-based access — analysts, admins, and auditors.

Customizable Reports:

Drag-and-drop export builder for tailored PDF / Excel outputs per stakeholder.

Platform Scalability

User Agent Parsing:

Parse user-agent strings to determine device types, OS, and browser usage for UX optimization.

Cloud-Native Infrastructure:

Kubernetes-based auto-scaling for Pig jobs with high-availability deployment on AWS / GCP / Azure.

ML Anomaly Detection:

Train classification models (Decision Tree, Isolation Forest) on log data to auto-flag security threats.

Microservices Architecture:

Decompose pipeline into independent services (ingest, parse, store, analyse) for improved maintainability.



Thank You

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Dept. of CSE (Data Science) | GRIET, Hyderabad | Academic Year 2026-27

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